Text Processing

Cleaning the text data

Unnecessary chars

Well this was fun! What do you think? 123#@!

HTML tags

I am going to Bangalore.

Emojis

He is 😳

Accented Text

Sómĕ Áccĕntĕd tĕxt

Real world data can be very messy

...

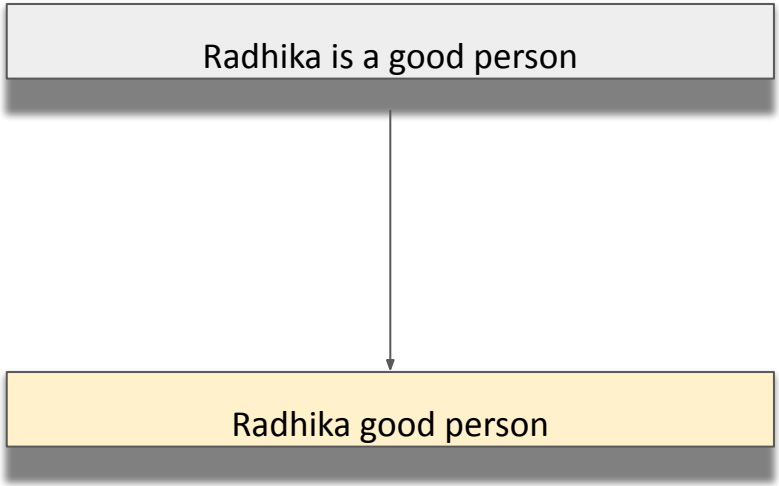
Exercise: Text Cleaning

You The

a and

1. High frequency words *i.e* present in most documents
2. Can not be used to distinguish between documents
3. Can be **removed** as features

Stopwords



Stopwords Removal

- For grammatical reasons, text documents are going to use different forms of a word...

Liking

Likes

Liked

Likely

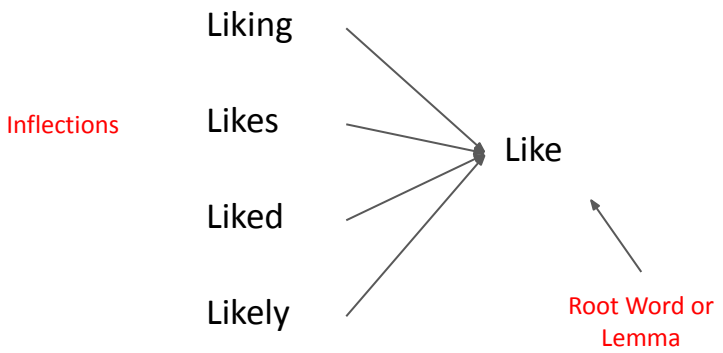
Inflections

It might be useful to consider only the base word for all the derived words...

How to change derived words into Base or **Root** word?

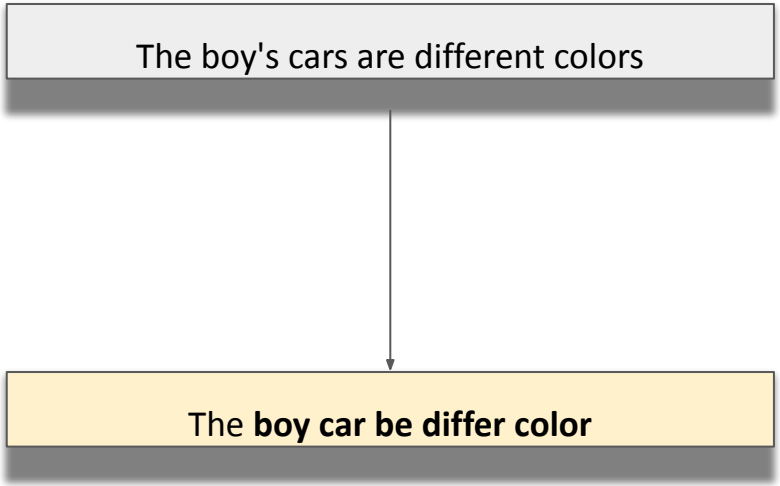
Two popular approaches …

1. Lemmatization
2. Stemming

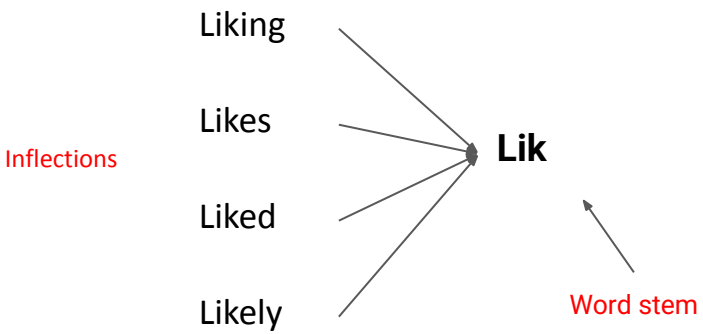


1. Converts inflections to root word or **Lemma**
2. Used for dimensionality reduction
3. Word stem usually **present in dictionary**

Lemmatization



Lemmatization



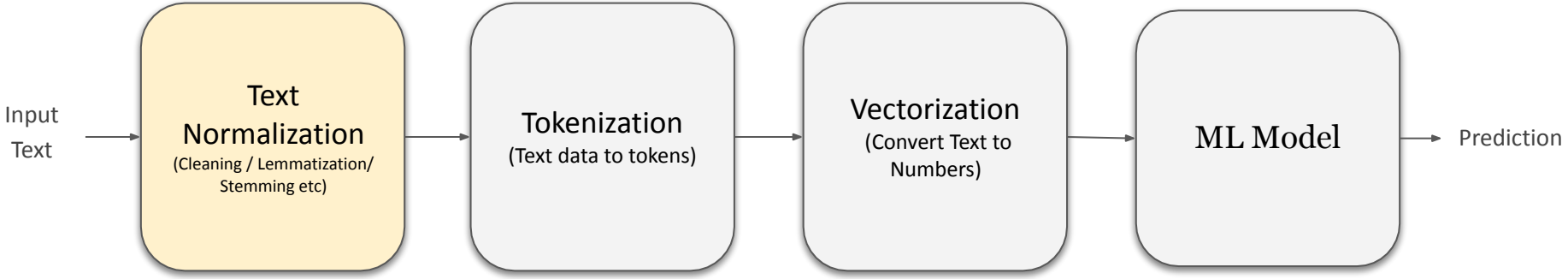
1. Converts inflections to root or word stem
2. Word stem may **not be present in dictionary**
3. Popular algorithms include Potter

   Stemmer, Lovins Stemmer etc

Stemming

Rule			Example		
SSES	→	SS	caresses	→	caress
IES	→	I	ponies	→	poni
SS	→	SS	caress	→	caress
S	→		cats	→	cat

Some Stemming rules



NLP Pipeline

Exercise

Text Normalization

Working with Web Data

Using Beautiful Soup (Exercise)

Information Retrieval

Barack Obama is an American politician who served as the 44th President of the United States from 2009 to 2017. He is the first African American to have served as president, as well as the first born outside the contiguous United States.

Extracting information

1. Who are the people that are being talked about in this paragraph?
2. What geography or locations are mentioned?
3. What time period is mentioned in this article?

Barack Obama is an American politician who served as the 44th President of the United States from 2009 to 2017. He is the first African American to have served as president, as well as the first born outside the contiguous United States.

What will it take to build an ML model to extract information?

Barack

Obama

is

an

American
politician

who

served

as

44th

President

of

the

United

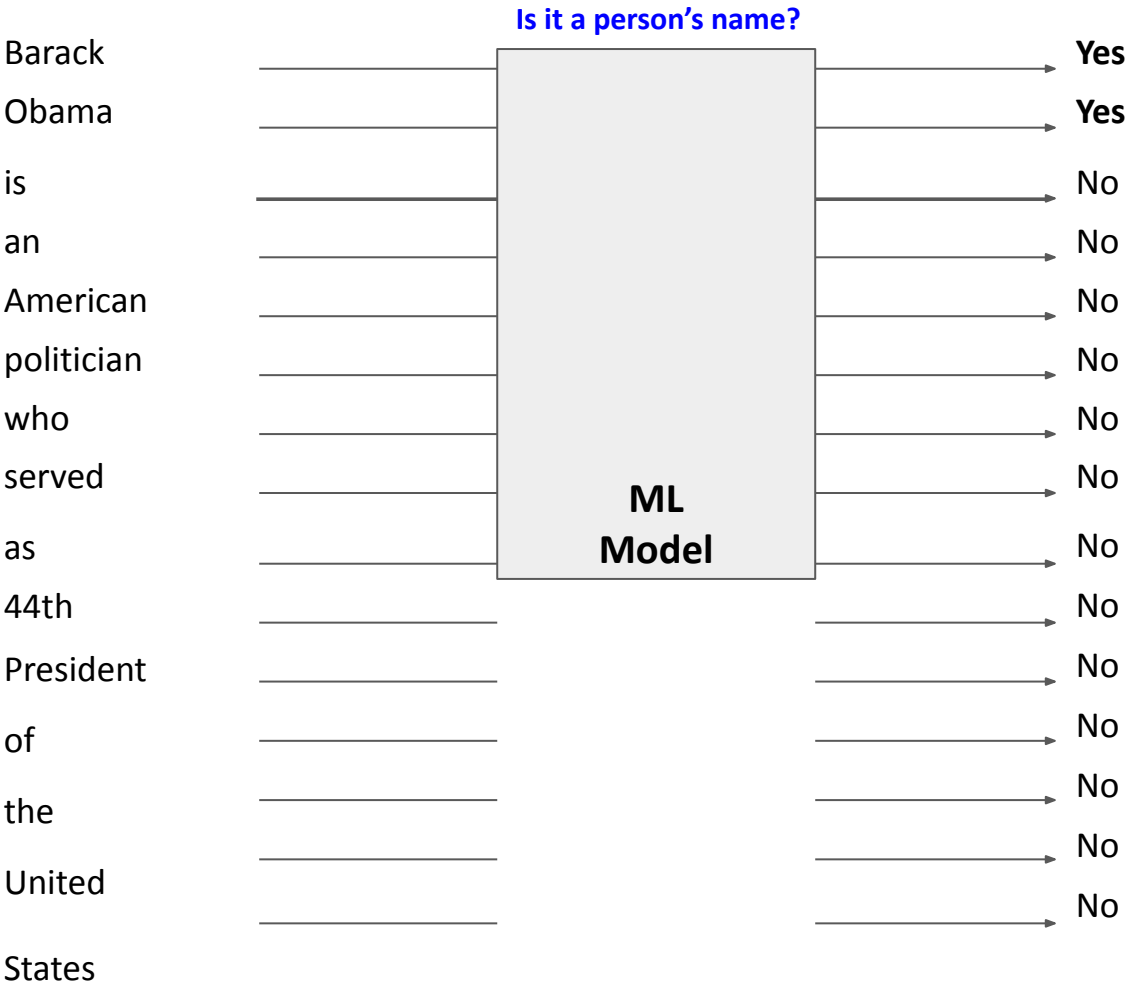
States

Our model needs to tell us

- if a word is a NAME

Is it a person's name?

Barack	**Yes**
Obama	**Yes**
is	No
an	No
American	No
politician	No
who	No
served	No
as	No
44th	No
President	No
of	No
the	No
United	No
States	No



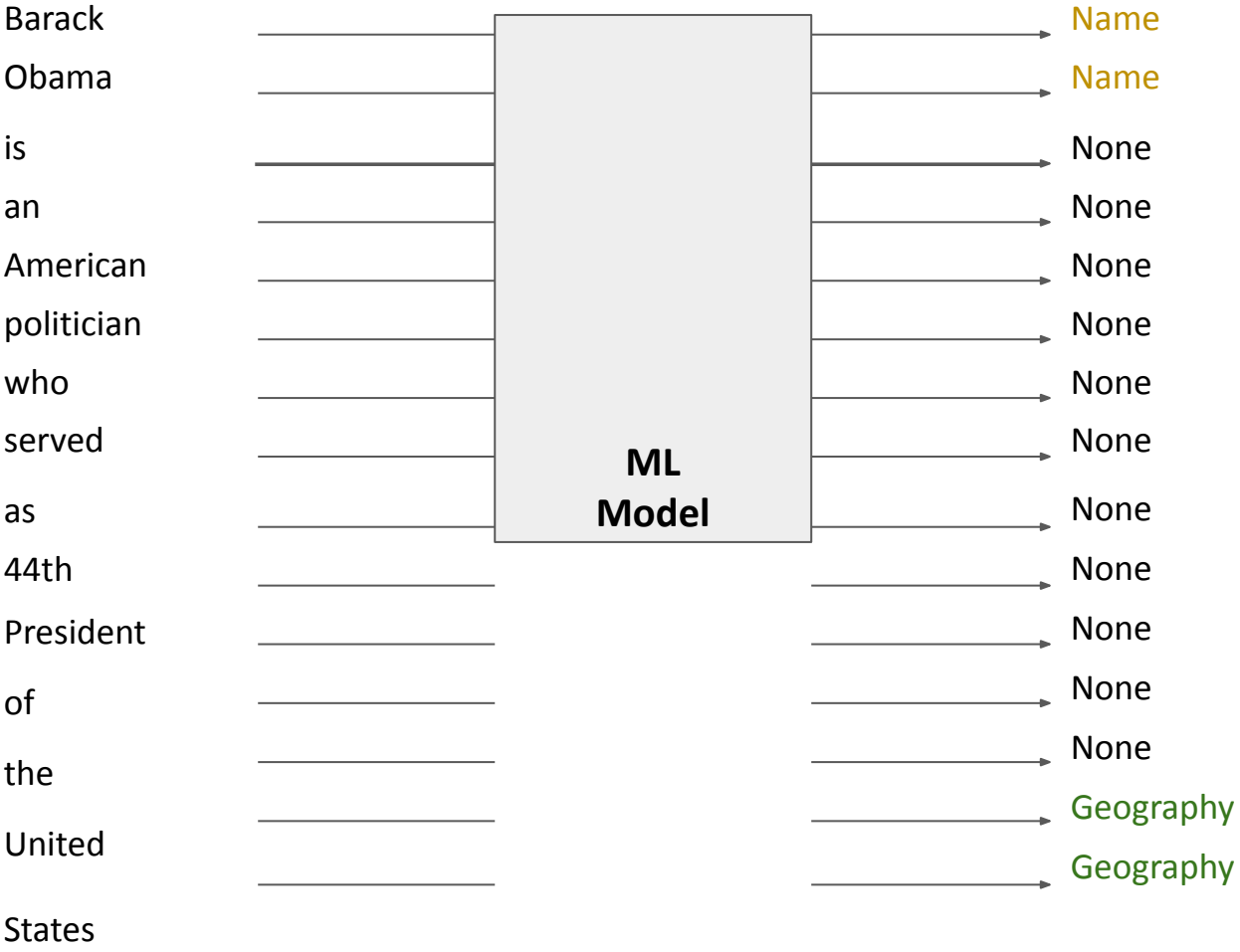
Classification or Regression?

- It's a classification model
- Word level classification
- Binary Classification

What if we want to know whether a word represents

1. A Geography or
2. Person’s name

Is it person's name, Geography or NONE?



What Class a word belongs to?

- Word level classification
- Multi label Classification
- 3 classes here
    - Person name
    - Geography
    - None

Barack Obama is an American politician who served as the 44th President of the United States from 2009 to 2017. He is the first African American to have served as president, as well as the first born outside the contiguous United States.

Named Entity Recognition Model

Classifies text into predefined categories or real world entities.

Named Entity Recognition Model

Some of the common labels/entities

TYPE	DESCRIPTION
PERSON	People, including fictional.
NORP	Nationalities or religious or political groups.
FAC	Buildings, airports, highways, bridges, etc.
ORG	Companies, agencies, institutions, etc.
GPE	Countries, cities, states.
LOC	Non-GPE locations, mountain ranges, bodies of water.
PRODUCT	Objects, vehicles, foods, etc. (Not services.)
EVENT	Named hurricanes, battles, wars, sports events, etc.
WORK_OF_ART	Titles of books, songs, etc.
LAW	Named documents made into laws.
LANGUAGE	Any named language.
DATE	Absolute or relative dates or periods.
TIME	Times smaller than a day.
PERCENT	Percentage, including "%".
MONEY	Monetary values, including unit.
QUANTITY	Measurements, as of weight or distance.
ORDINAL	"first", "second", etc.
CARDINAL	Numerals that do not fall under another type.

Barack Obama **PERSON** is an American **NORP** politician who served as the 44th **ORDINAL** President of the United States **GPE** from 2009 to 2017 **DATE**. He is the first **ORDINAL** African American **NORP** to have served as president, as well as the first **ORDINAL** born outside the contiguous United States **GPE**.

Named Entity Recognition (NER) model's output

Good News!!!

There are lots of pre-trained Named Entity Recognition models are available which we can use out of the box.

Where to use NER models

1. Information extraction
2. Improving search results
3. Content Recommendations
4. … many more

She saw a **bear.**

Your efforts will **bear** fruit.

How the word ‘bear’ different in two sentences?

Noun

She saw a **bear.**

Verb

Your efforts will **bear** fruit.

Words play different roles in a sentence.

How do we find extract role a word plays in a sentence?

Part of Speech (PoS) tagger

Part-of-Speech (POS) Tagging

Hyderabad	is	the	capital	of	Telangana.
PROPN	VERB	DET	NOUN	ADP	PROPN

1. Assign grammatical properties (e.g. noun, verb, adverb, adjective etc.) to words.
2. These properties can used to extract information by building language rules.
3. Multiple NLP libraries support POS tagging e.g. NLTK, spaCy

Part-of-Speech (POS) Tagging

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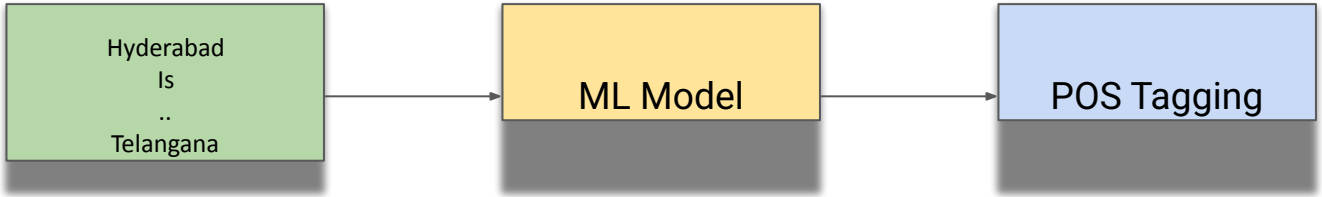

Pre-trained POS tagging model are available in several libraries

PoS Tagging, Named Entity Recognition

Using spaCy

What is the key to building models like NER, PoS tagger etc?

Obama is an American politician who served as the 44th President of the United States from 2009 to 2017.

While in school, I played a game called **Obama**.

What makes a name … **a name**?

Obama is an American politician who served as the 44th President of the United States from 2009 to 2017.

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What makes a name … **a name**?

It's context…

word's neighbours and sequence of words

I had washed my house.

I had my house washed.

Exact same words.. but different meaning.

Why?

Document #1

I had washed my house.

Document #2

I had my house washed.

Vocabulary

had, house, I, my, washed

	had	house	I	my	washed
Index	0	1	2	3	4
Document #1	1	1	1	1	1
Document #2	1	1	1	1	1

CountVectorizer can not help distinguish these two examples

Problem

Bag of words, CountVectorizer, Tf-IDF …

1. does not keep order of words
2. can not be used to when understanding **context** is important.

Solution?

1. One hot encoding
2. Word2Vec